

Linear Scaling Causal Discovery from High-Dimensional Time Series by Dynamical Community Detection

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Understanding which parts of a dynamical system cause each other is extremely relevant in fundamental and applied sciences. However, inferring causal links from observational data, namely, without direct manipulations of the system, is still computationally challenging, especially if the data are high dimensional. In this Letter we introduce a framework for constructing causal graphs from high-dimensional time series, whose computational cost scales linearly with the number of variables. The approach is based on the automatic identification of *dynamical communities*, groups of variables which mutually influence each other and can therefore be described as a single node in a causal graph. These communities are efficiently identified by optimizing the information imbalance, a statistical quantity that assigns a weight to each putative causal variable based on its information content relative to a target variable. The communities are then ordered starting from the fully autonomous ones, whose evolution is independent from all the others, to those that are progressively dependent on other communities, building in this manner a community causal graph. We demonstrate the computational efficiency and the accuracy of our approach on discrete-time and continuous-time dynamical systems including up to 80 variables.

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Introduction—The growing abundance of time series—spanning fields from environmental monitoring to finance and neuroscience—offers unprecedented opportunities for understanding real world observations. Specifically, causal discovery allows studying how different parts of a system influence each other and inferring the existence of directional couplings (causal relationships) between variables in a time series [1–3].

Relationships between variables can be depicted, using Pearl's approach [4,5], as a time series graph. Each dynamic variable at a specific time is represented by a node and an arrow from one node to another represents a direct causal link. Such a graph encodes all conditional independence relationships between pairs of variables [2,6,7]. Several strategies can be used to build this graph. A first option involves checking for each couple of lagged variables if there is no conditioning set that makes them independent. An arrow between the two nodes is then drawn only if such set does not exist [8]. A strategy to assess this condition is to employ an iterative approach, where the size of the tested set is progressively increased [9,10]. Although this approach leads to optimal detection power, it requires performing a number of tests that scales exponentially with the number of variables. Therefore, strategies to reduce the search space in practical

applications have been developed [11,12]. Alternatively, one can consider for each pair of lagged variables the largest possible conditioning set, which includes all past history of the time series. The multivariate versions of Granger causality [13–15] and transfer entropy [16] can be regarded as implementations of this second strategy [2,8]. Although this approach drastically reduces the number of tests to be performed, requiring a single test per pair of variables, it is significantly affected by the curse of dimensionality, often resulting in low detection power [11].

In this Letter, we introduce a method for causal discovery designed for high-dimensional time series. A powerful feature of our approach is that it enables the identification of causal influences that emerge from the collective dynamics of multiple variables at a very moderate computational cost, which, importantly, scales linearly with the number of variables. The graph obtained with our algorithm is a “mesoscopic” version of the standard graphs, as it groups together variables whose evolution cannot be described independently. We call these groups of variables *dynamical communities*. Our approach aims at revealing, if present, a hierarchical organization of these communities, emerging from unidirectional inter-group interactions. We refer to the resulting graph as *community causal graph*.

The key ingredient of our method is the efficient identification of the dynamical communities. This is achieved by optimizing the information imbalance [17,18] as a function of a set of variational weights, one for each variable of the system, according to a prediction criterion

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broadly inspired by Granger causality [14,15]. We will show that the values of the weights allow identifying efficiently the communities and, as a consequence, building a causal graph, avoiding combinatorial searches of conditioning sets.

We build the method on the assumption of causal sufficiency, namely that there are no unobserved common drivers of two or more dynamic variables. In Supplemental Material (SM) [19] we discuss the effect of unobserved variables in different scenarios. We demonstrate the effectiveness of our approach using time series generated from both discrete-time and continuous dynamical systems, showcasing its applicability to high-dimensional scenarios.

The algorithm exploits the information imbalance [17], which allows quantifying the information content of different distance measures defined on a data set. The underlying idea is that a distance measure d^A is predictive with respect to another distance measure d^B if points close according to d_A are also close according to d^B . The information imbalance is defined as

$$\Delta(d^A \rightarrow d^B) = \frac{2}{N^2} \sum_{\substack{i,j \\ (i \neq j)}} \delta_{r_{ij}^A, 1} r_{ij}^B, \quad (1)$$

where N is the number of points in the dataset, δ is the Kronecker delta and $r_{ij}^{(\cdot)}$ is the distance rank of point j with respect to point i . The superscript refers to the distance used in the computation. For example, $r_{ij}^A = 2$ if j is the second nearest neighbor of i according to distance d^A . Equation (1) defines a quantity that, in the limit of large N , approaches zero when all nearest neighbors in space A remain nearest neighbors in space B , namely, when d^A is maximally predictive of d^B .

In Ref. [24] we showed that the information imbalance can be used to infer the presence of causality between two multidimensional dynamical systems X and Y . We assumed that if X causes Y and one attempts to make a prediction of the future of Y , a distance measure built using the present states of both X and Y will have more predictive power than a distance built using only Y . Formally, we assumed that X causes Y if and only if

$$\hat{w} = \underset{w}{\operatorname{argmin}} \Delta(d^{wX(0), Y(0)} \rightarrow d^{Y(\tau)}) \neq 0 \quad (2)$$

for some positive time lag τ . The notation $d^{(\cdot)}$ denotes the squared Euclidean distance built over the superscript variables. For example,

$$d_{ij}^{wX(0), Y(0)} = w^2 \|X_i(0) - X_j(0)\|^2 + \|Y_i(0) - Y_j(0)\|^2, \quad (3)$$

where the Latin letters i and j denote independent realizations of the same dynamics, obtained either from

uncorrelated samples of a single stationary trajectory, or from distinct trajectories with independent initial conditions. In SM [19] we relate the criterion in Eq. (2) to the notion of conditional independence. Importantly, the approach in Ref. [24] builds on prior knowledge of the groups of variables that make up the distinct dynamical systems interacting with each other (the dynamical communities in the language of this Letter, see below). This is a very strong assumption, which for real-world data is typically violated. This Letter is dedicated to overcoming this problem.

Here, we extend the approach described above to automatically and efficiently find those dynamical communities. Our algorithm makes use of a differentiable extension of the information imbalance (DII) [18]:

$$\text{DII}(d^A \rightarrow d^B) = \frac{2}{N^2} \sum_{\substack{i,j \\ (i \neq j)}} \frac{e^{-d_{ij}^A/\lambda}}{\sum_{m(\neq i)} e^{-d_{im}^A/\lambda}} r_{ij}^B. \quad (4)$$

We note that Eq. (4) tends to Eq. (1) in the limit $\lambda \rightarrow 0$. If distance d^A depends on a set of parameters $\mathbf{w}$, this formulation allows optimizing such parameters by gradient descent.

Our algorithm, illustrated in Fig. 1, can be conceptually divided into three parts: (i) the DII is minimized to infer each variable's *autonomous set*, namely the set of variables that directly or indirectly cause it; (ii) the autonomous sets are used to find the dynamical communities, namely sets of variables which directly or indirectly influence each other; and (iii) a macroscopic graph depicting the causal interactions between the communities is constructed. The input to the algorithm is a set of D time-dependent variables $\{X^\alpha(t)\}_{\alpha=1}^D$. We will refer to such variables as “microscopic,” in contrast to the “mesoscopic” dynamical communities identified by the method.

Identification of autonomous sets—As a first step we infer the *autonomous set* $\mathcal{S}^\beta$ of each variable X^β , which we define as the set of all variables $\{X^\alpha\}$ directly or indirectly causing X^β . X^α is a direct cause of X^β if, for some time lag τ , there exists a direct link $X^\alpha(0) \rightarrow X^\beta(\tau)$ in the ground-truth time series graph. Conversely, X^α is an indirect cause of X^β if, for any $\tau > 0$, the directed paths connecting $X^\alpha(0)$ and $X^\beta(\tau)$ pass through at least a third variable $X^\gamma(\tau')$ ($\gamma \neq \alpha, \gamma \neq \beta, 0 < \tau' < \tau$). We will use the notations $X^\alpha \rightarrow X^\beta$ and $X^\alpha \in \mathcal{S}^\beta$ interchangeably, without distinguishing between direct and indirect links among the microscopic variables.

To identify the autonomous sets $\mathcal{S}^\beta$, we optimize the information imbalance of Eq. (4) between a distance measure built with all dynamic variables at time $t = 0$ and a distance built with a single variable X^β at time $t = \tau$. To level out the fluctuation ranges of different variables, we first scale each variable by its standard deviation over

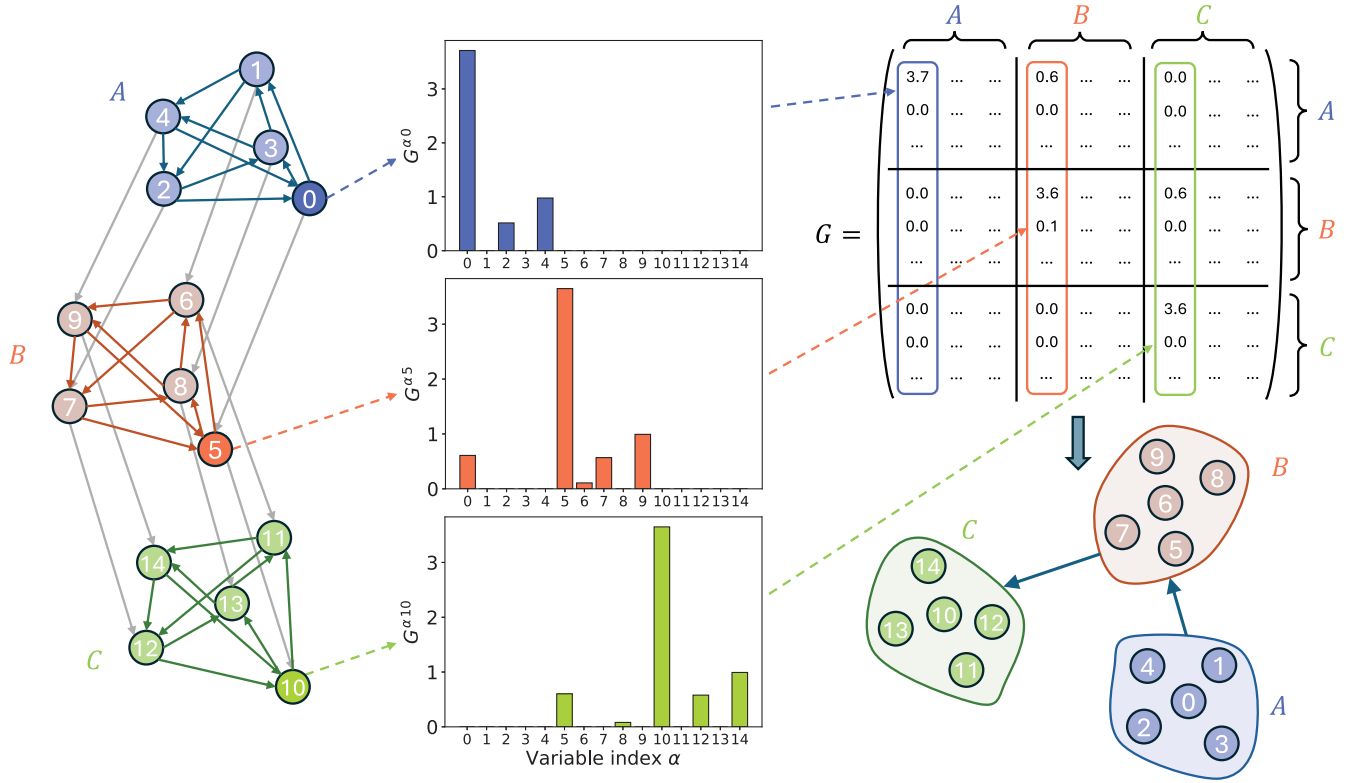


FIG. 1. Illustration of the algorithm, using a 15-dimensional dynamical system composed of three groups of noisy coupled logistic maps. On the left, the ground-truth connectivity of the system is depicted in an all-variable representation. Repeated optimizations of DII ($d^{w \odot X(0)} \rightarrow d^{X^\beta(\tau)}$) are carried out for each target variable β according to Eq. (5) ($\beta = 0, 5$, and 10 are shown in blue, red, and green, respectively). The optimal weights from each optimization, depicted in the central bar plots, are used to construct a connectivity matrix G (here, G is built using a single time lag $\tau = 1$). From this matrix, dynamical communities are identified and depicted on a graph, where several variables are grouped within the same node.

the entire trajectory. Thus, the weights obtained by this optimization are

$$\hat{w}_\beta = \underset{w}{\operatorname{argmin}} \operatorname{DII} \left(d^{w \odot X(0)} \rightarrow d^{X^\beta(\tau)} \right), \quad (5)$$

where $\odot$ denotes the element-wise product. We generalize the principle of Eq. (2) by stating that X^α is a direct or indirect cause of X^β when the α component of $\hat{w}_\beta$, denoted by $\hat{w}_{\beta}^\alpha$, is nonzero.

Following the intuition that different couplings might manifest at different timescales [24–26] we repeat the optimization in Eq. (5) for several values of τ between 1 and $\tau_{\max}$, where $\tau_{\max}$ is a hyperparameter which, in applications, we take of the order of the autocorrelation time of X^β . As depicted in Fig. 1, the maximum weights over the tested values of τ are stored as columns of a $D \times D$ matrix G :

$$G^{\alpha\beta} = \max_{\tau} \hat{w}_{\beta}^\alpha. \quad (6)$$

Constructing d^A over a single time frame provides the correct results when the lag of direct links is not larger

than 1 in the underlying time series graph. In SM [19] we describe how our approach can be extended when this condition does not hold, constructing d^A on multiple time frames.

Ideally, each autonomous set $\mathcal{S}^\beta$ can be directly extracted from the nonzero elements of $G^{\alpha\beta}$. However, in applications G is estimated using a finite number of measures (or a finite trajectory). Therefore, one can decide if an element is zero only according to a specified tolerance: we then set $X^\alpha \in \mathcal{S}^\beta$ whenever $G^{\alpha\beta} > \varepsilon$. The threshold ε is the main hyperparameter of our algorithm. A false negative may appear if the coupling that we aim to detect by testing $\hat{w}_{\beta}^\alpha > 0$ is present but weak. This may occur, for example, when X^α causes X^β indirectly via several mediating variables.

To account for false negatives, we first construct a directed graph represented by the set of links for which $G^{\alpha\beta} > \varepsilon$, and then we construct each autonomous set $\mathcal{S}^\beta$ as the full set of ancestors of X^β . Missing links $X^\alpha \rightarrow X^\beta$ are likely to be recovered through indirect paths $X^\alpha \rightarrow X^\gamma \rightarrow \dots \rightarrow X^\beta$.

Identification of dynamical communities—In the second step, hierarchical structure of dependencies between groups

of variables can be directly retrieved from the analysis of the sets $\mathcal{S}^\beta$.

In particular, given an autonomous set $\mathcal{S}^\beta$, we will define it as *minimal* if, for every variable $x^\alpha \in \mathcal{S}^\beta$, one has $\mathcal{S}^\alpha \equiv \mathcal{S}^\beta$. In a minimal autonomous set, each variable depends on all and only the other variables in the set: together they form a *dynamical community* $\mathcal{G}_k$ whose evolution is independent of the rest of the network. Once all sets $\mathcal{G}_k$ have been identified, their variables can be removed from the graph, and new sets with the same property can be extracted. This will reveal new dynamical communities, which we distinguish from the initial ones by assigning them a higher autonomy level. By convention, we assign autonomy level 0 to the first dynamical communities identified, and we increase such a level by 1 at each iteration. Notice that if a

group has level of autonomy k , then it can be caused only by groups with level $m < k$, and, in turn, it can cause only groups with level $m > k$; no links can be present among groups at the same level.

Construction of the community causal graph—In the third step, the causal connections between the dynamical communities are depicted using a directed acyclic graph where each node represents a dynamical community $\mathcal{G}_k$, which we name *community causal graph* (see Fig. 1). We draw a directed arrow $\mathcal{G}_k \rightarrow \mathcal{G}_m$ if there is at least a pair of variables $X^\alpha \in \mathcal{G}_k$ and $X^\beta \in \mathcal{G}_m$ such that $X^\alpha \rightarrow X^\beta$. Directed arrows $\mathcal{G}_k \rightarrow \mathcal{G}_m$ are omitted whenever $\mathcal{G}_k$ and $\mathcal{G}_m$ are also connected through an indirect path $\mathcal{G}_k \rightarrow \dots \rightarrow \mathcal{G}_m$.

By construction, links that connect communities with consecutive levels of autonomy are direct (e.g., $\mathcal{G}^k \rightarrow \mathcal{G}^m$),

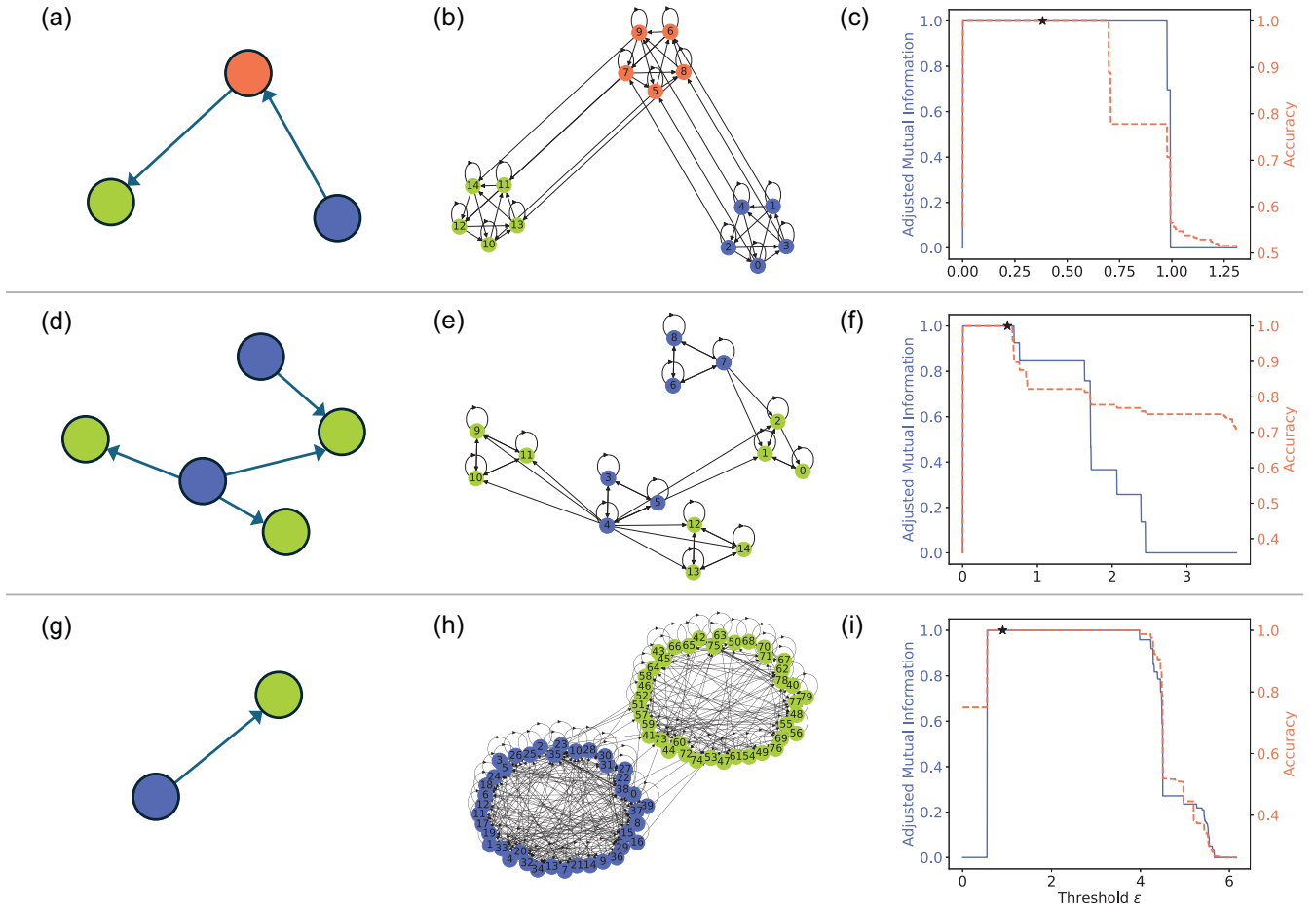


FIG. 2. Outcome and performance of the algorithm in three different test cases: (a)–(c) 15 coupled logistic maps, (d)–(f) five coupled Lorenz systems, and (g)–(i) two Lorenz 96 systems. Left panels (a) and (d) and (g): community causal graphs produced by the algorithm, which correctly retrieves the ground-truth connectivity. Central panels (b) and (e) and (h): all-variable graphs obtained from condition $G^{\alpha\beta} > \epsilon$, setting ϵ to 0.38, 0.6 and 3.5, respectively. Nodes in community and all-variable graphs are colored according to the autonomous levels identified in step ii). Right panels (c) and (f) and (i): validation measures of the recovered connectivity as a function of the threshold parameter ϵ , in the range $[0, \max_{\alpha\beta(\alpha \neq \beta)} G^{\alpha\beta}]$. For larger values of ϵ , no pairs of variables are found to be linked. Blue solid curve: adjusted mutual information between the retrieved groups and the ground-truth ones. Red dashed curve: accuracy of the all-variable adjacency matrix recovered from the final community graph, defined as the fraction of correctly identified links. Black star marker: results by setting the threshold ϵ to the average of all weights in $G^{\alpha\beta}$.

whereas links between nonconsecutive communities may be indirect. In Appendix A, we show that one can distinguish between direct and indirect links among nonconsecutive communities by a simple *a posteriori* analysis.

Results—We tested our approach on trajectories generated by dynamical systems of different complexity. Specifically, we considered three groups of five coupled logistic maps [Figs. 2(a)–2(c)], five coupled Lorenz oscillators [27] [Figs. 2(d)–2(f)], and two 40-dimensional Lorenz 96 systems [28] unidirectionally coupled [Figs. 2(g)–2(i)]. Since deterministic relationships are known to violate a condition known as faithfulness [2,8] we added a small white noise to each variable while integrating the dynamic equations (see Appendix B). We report in SM [19] additional tests in presence of observational noise. For each system, distances entering the DII optimization at step i) of the algorithm were computed by extracting $N = 2000$ frames, used as independent initial conditions, from a single time series realization.

The left panels in Fig. 2 show the community graphs produced by the algorithm when the threshold ϵ is set to the average value of the weights in matrix G (black star marker in the right panels). These graphs reproduce the correct connectivity read from the ground-truth equations: all variables grouped in a single node are dynamically intertwined; namely, each one is a direct or indirect cause of any other variable, and an arrow between two groups at consecutive levels is present when at least two variables, one per group, are interacting. The central panels show the all-variable graphs built from the matrix $G^{\alpha\beta} > \epsilon$, with nodes colored according to the level of autonomy assigned to the corresponding groups in the second step of the algorithm. This matrix was computed according to Eq. (6), with $\tau_{\max} = 1$ for the top row example, $\tau_{\max} = 60$ for the middle row, and $\tau_{\max} = 30$ for the bottom row example.

The outcome of the algorithm is influenced by the choice of the threshold ϵ . In Fig. 2, right column, we validated the correctness of the recovered dynamical communities and their connectivity separately, by measuring as a function of ϵ the adjusted mutual information [29] between the true groups and the recollected ones (right panels, blue solid curves), and the accuracy of the all-variable adjacency matrix obtained from the final community graphs (right panels, dashed red curves). Both measures are defined in the range $[0, 1]$, with 1 being the case of optimal reconstruction. Refer to SM [19] for a formal definition of these measures. Remarkably, in all systems we observe a wide range of threshold values leading to an exact reconstruction, and a false positive rate which is nonzero only for negligibly small values of the threshold ($\epsilon < 10^{-16}$, $\epsilon < 10^{-13}$ and $\epsilon < 0.55$ for the systems in panels (a), (d), and (g), respectively).

Discussion—We have demonstrated that our approach can efficiently reconstruct the causal structure underlying high-dimensional dynamical systems, providing a coarse-grained visualization of the system’s causal connectivity,

which has been a topic of growing interest [30]. The most relevant feature of our method is that the number of optimizations—analogue to conditional independence tests—scales linearly with the number of variables, and still the approach is able to capture “multibody” synergistic causal effects that are hard to detect by constructing conditioning sets of increasing size [2].

For comparison, we applied a state-of-the-art method for causal discovery on time series data, PCMCi [11], to the coupled Lorenz 96 systems. This algorithm efficiently reduces the search space for conditioning sets, at the cost of introducing some hyperparameters. In the SM, we show that the two 40-dimensional communities are correctly identified for certain hyperparameter combinations. However, the link among the communities appears hard to detect, as it results from few intercommunity connections.

We notice one could consider analyzing the G matrix with graph clustering methods [31], rather than using it to identify communities. One could also employ this matrix to speed up the search of relevant groups of variables according to other criteria [32–35], even in presence of bidirectional couplings. The approach can be improved by estimating the statistical confidence of each element in G , allowing the application of confidence-based thresholds, instead of a fixed ϵ . We consider the usage of multiple time delays τ in Eq. (6) as a valuable feature of our approach. In SM (Fig. S2) [19] we show that using only $\tau = 1$ would significantly decrease the detection power of the algorithm.

The codes implementing our algorithm are available in the Python library DADAPy [36].

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Data availability—The data that support the findings of this article are openly available [37].

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Correction: In the section entitled “Construction of the community causal graph,” at the end of the first paragraph, the description of the procedure implemented in the code has been modified.

End Matter

Appendix A: Community graph refinement—To properly interpret the arrows of the community graph, we define a link between two dynamical communities $\mathcal{A}$ and $\mathcal{B}$ as “direct” if there is at least a pair of variables, $X^\alpha \in \mathcal{A}$ and $X^\beta \in \mathcal{B}$, such that X^α is a direct cause of X^β . In contrast, we call such a link “indirect” if *all* causal paths between their constituent microscopic variables are indirect. Links that connect communities with consecutive levels of autonomy are necessarily direct, as the absence of “mediating” communities implies the existence of at least a pair of microscopic variables, one for each group, that are directly linked. On the other hand, communities $\mathcal{A}$ and $\mathcal{B}$ in the pattern $\mathcal{A} \rightarrow \mathcal{C} \rightarrow \mathcal{B}$ may be directly linked, although such a link does not correspond to any direct arrow in the community graph.

Whenever a pattern $\mathcal{A} \rightarrow \mathcal{C} \rightarrow \mathcal{B}$ appears in the community graph, the existence of a direct link from $\mathcal{A}$ to $\mathcal{B}$ can be assessed by treating each community as single multi-dimensional variable. In the following, $\mathcal{C}$ may possibly represent a sequence of mediating groups. A direct link $\mathcal{A} \rightarrow \mathcal{B}$ is present if there exists a pair of microscopic variables $X^\alpha \in \mathcal{A}$ and $X^\beta \in \mathcal{B}$ that are directly linked. We note that this definition of inter-group link is consistent with that provided within the general causal discovery framework of ref. [30]. Assuming the absence of instantaneous interactions, this condition is fulfilled if there exists a lag τ for which $\mathcal{B}(\tau)$ depends on $\mathcal{A}(0)$, conditioning over

$$\{\mathcal{B}(\tau-1), \dots, \mathcal{B}(\tau-E), \mathcal{C}(\tau-1), \dots, \mathcal{C}(\tau-E)\}. \quad (\text{A1})$$

Here, E represents the maximum lag of the microscopic direct links. Conditioning over the driven ($\mathcal{B}$) and mediating ($\mathcal{C}$) communities in the E frames preceding $t = \tau$ ensures that all microscopic paths from $\mathcal{A}(0)$ to $\mathcal{B}(\tau)$ are blocked in the sense of d separation [8].

Importantly, τ does not represent the specific lag of the putative direct link. As commented in the Discussion, the value of τ that maximizes this conditional dependence in practice may be significantly larger than the maximum lag of the direct microscopic interactions.

The above conditional independence test can be translated into the minimization of the DII, similarly to Eq. (5). In the case $E = 1$, this minimization reads,

$$\min_{\mathbf{w}} \text{DII}(d^{\mathbf{w} \odot [\mathcal{A}(0), \mathcal{B}(\tau-1), \mathcal{C}(\tau-1)]} \rightarrow d^{\mathcal{B}(\tau)}), \quad (\text{A2})$$

where $\mathbf{w} \odot [\cdot]$ denotes a “group-wise” product such that, for each community in the list, all its variables are scaled by a single weight (if $\mathcal{C}$ is a single community, $\mathbf{w}$ has four components). If the optimal weight associated to $\mathcal{A}(0)$ is different from zero for some lag τ , we conclude that a direct

link $\mathcal{A} \rightarrow \mathcal{B}$ exists, and we draw the corresponding arrow in the refined community graph.

The number of optimizations to be performed to test all pairs of linked and nonconsecutive communities depends on the topology of the community graph. In the worst-scaling scenario, which is the case of a nonbranched chain of D' communities ($\mathcal{G}^0 \rightarrow \mathcal{G}^1 \rightarrow \dots \rightarrow \mathcal{G}^{D'}$), the number of tests is equal to $(D' - 1)(D' - 2)/2$. This number comes by counting $D' - 2 - k$ optimizations for each community $\mathcal{G}^k$, where $\mathcal{G}^k$ is the putative “driver” community (except for the last two communities, which do not have nonconsecutive groups). We stress that this scaling is quadratic in the number of dynamical communities, which is expected to be substantially smaller than the number of microscopic variables. Moreover, optimizations such as that of Eq. (A2) can be skipped whenever the all-variable adjacency matrix constructed from matrix $G^{\alpha\beta}$ (step i of the algorithm) does not display any link between variables in $\mathcal{A}$ and variables in $\mathcal{B}$, which is a necessary condition to have a direct link from $\mathcal{A}$ to $\mathcal{B}$.

In SM [19] we validate this approach on two systems of coupled logistic maps, showing that this test allows for a consistent refinement of the community where direct links between nonconsecutive communities are explicitly represented.

Appendix B: Details on test systems—In this section we provide details on the test systems employed in the validation tests. To write the equations in a compact form, we use indices μ, ν to identify different dynamical communities and indices α, β to represent variables within the same community. Trajectories of 10^5 time frames were generated for all systems. The first 10^4 samples of each trajectory were then discarded to eliminate equilibration artifacts.

Logistic maps—The equations of the noisy coupled logistic maps shown in Figs. 1 and 2(a) of the main text, structured in three communities with five variables each ($\mu, \nu \in \{0, 1, 2\}$; $\alpha, \beta \in \{0, 1, 2, 3, 4\}$), read ,

$$X_\mu^\alpha(t+1) = X_\mu^\alpha(t) \left(r_\mu^\alpha - r_\mu^\alpha X_\mu^\alpha(t) - \sum_{\beta=0}^4 c_\mu^{\beta\alpha} X_\mu^\beta(t) \right. \\ \left. - \sum_{\nu=0}^2 d_{\nu\mu}^\alpha X_\nu^\alpha(t) + \sigma \mathcal{R}_\mu^\alpha(t) \right) \mod 1. \quad (\text{B1})$$

The terms $\sigma \mathcal{R}_\mu^\alpha(t)$, where $\sigma = 0.1$ and $\mathcal{R}_\mu^\alpha \sim \mathcal{N}(0, 1)$, are independent white noises added to all variables of the system. The coefficients $c_\mu^{\beta\alpha}$ tune the strength of the interactions within the same community, while the

couplings $d_{\mu\nu}^\alpha$ control the interactions between different communities. For each α , we chose the parameters r_μ^α by sampling uniformly 10 values in (3.68,4), and setting $r_0^\alpha = r_1^\alpha \neq r_2^\alpha$. Within each community μ we considered $c_{\mu}^{\beta\alpha} = \delta_{\beta,\alpha-1} + 0.5\delta_{\beta,\alpha+2}$, with conventions $x_\mu^{-1} = x_\mu^4$ and $x_\mu^5 = x_\mu^0$, and for each variable α in community μ we set the interaction with variables of other communities as $d_{\nu\mu}^\alpha = 0.5(\delta_{\nu,0}\delta_{\mu,1} + \delta_{\nu,1}\delta_{\mu,2})$. The connectivity of the system is depicted in Fig. 1 of the main text.

The complementary system displaying a direct link from the communities $\mu = 0$ and $\mu = 2$ (Fig. S5 in SM [19]) was constructed by considering an additional term in the inter-community interaction: $d_{\nu\mu}^\alpha = 0.5(\delta_{\nu,0}\delta_{\mu,1} + \delta_{\nu,1}\delta_{\mu,2} + \delta_{\nu,0}\delta_{\mu,2})$.

Lorenz systems—The system of five coupled Lorenz systems ($\mu, \nu \in \{0, 1, 2, 3, 4\}$) is described by the following Itô stochastic differential equations, reported here for a single system (or community) μ :

$$\begin{cases} dX_\mu^0 = 10(X_\mu^1 - X_\mu^0)dt + dW_\mu^0 \\ dX_\mu^1 = (X_\mu^0(28 - X_\mu^2) - X_\mu^1 + \\ \quad + c \sum_{\nu=0}^4 d_{\nu\mu}(X_\nu^0)^2)dt + dW_\mu^1 \\ dX_\mu^2 = (X_\mu^0 X_\mu^1 - \frac{8}{3}X_\mu^2)dt + dW_\mu^2 \end{cases} \quad (\text{B2})$$

The coupling strength was fixed to $c = 0.3$ and $d_{\nu\mu}$, which defines the interaction topology among communities [Fig. 2(e) of the main text] was set to

$$d = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{B3})$$

Equation (B2) were integrated using the Euler-Maruyama algorithm for Itô equations. The trajectory was calculated with a sampling time of $\delta t = 0.003$. The noise was taken independent for each variable at each time step [autocorrelation $A(\tau) = \delta(\tau)$].

Lorenz 96 systems—The two unidirectionally coupled Lorenz 96 systems of 40 variables each ($\mu, \nu \in \{0, 1\}$, $\alpha, \beta \in \{0, 1, \dots, 39\}$) are defined by the following ordinary

differential equations:

$$dX_\mu^\alpha = \left((X_\mu^{\alpha+1} - X_\mu^{\alpha-2})X_\mu^{\alpha-1} - X_\mu^\alpha + F_\mu + c\delta_{\mu,1}X_0^\alpha \right) dt + dW_\mu^\alpha, \quad (\text{B4})$$

where $X_\mu^{-1} = X_\mu^{39}$, $X_\mu^{40} = X_\mu^0$, $F_0 = 5$, $F_1 = 6$, and $c = 0.75$. Equation (B4) were integrated with time step $dt = 0.03$ using uncorrelated noise and the Euler-Maruyama algorithm as above.

In the Lorenz and Lorenz 96 systems, the coupling terms among communities have the same functional form employed in ref. [24].

Appendix C: Validation measures—The adjusted mutual information (AMI) [29] shown in the right panels of Fig. 2 is a measure of discrepancy between the dynamical communities retrieved by our algorithm, $\{\mathcal{G}_i\}$, and the ground-truth groups $\{\mathcal{G}_i^{gt}\}$. The sets $U := \{\mathcal{G}_i\}$ and $V := \{\mathcal{G}_i^{gt}\}$ define two possible partitions of the D dynamical variables. We computed the AMI by using the `metrics.adjusted_mutual_info_score` function in SciPy [38], which computes it as

$$\text{AMI}(U, V) = \frac{I(U, V) - \mathbb{E}[I(U, V)]}{(H(U) + H(V))/2 - \mathbb{E}[I(U, V)]}, \quad (\text{C1})$$

where $I(U, V)$ is the mutual information between the two partitions, $H(U)[H(V)]$ is the Shannon entropy associated to partition U (respectively, V), and $\mathbb{E}[I(U, V)]$ is the expected mutual information between two random partitions.

To measure the agreement of the retrieved links among dynamical communities with the ground-truth connectivity, we first constructed from the final community graph an all-variable adjacency matrix embedding all direct and indirect links retrieved by the algorithm. Then, we computed the accuracy of this connectivity matrix as the fraction of correctly retrieved links over the total number of links:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{P} + \text{N}}, \quad (\text{C2})$$

where TP (TN) is the number of true positive (negative) link detections, and P (N) is the total number of positive (negative) detections.